



Airplane Scatter Plugin

Quick Reference · FM-DX-Webserver · Author: Highpoint

v2.2 · April 2026

✈️ WHAT IS AIRPLANE SCATTER?

An aircraft at **8 000–12 000 m cruise altitude** reflects FM signals between a distant transmitter and your receiver — even when both are beyond each other's radio horizon.

The plugin **predicts these events in real time**, up to 3 min ahead, and displays them on an interactive map with a live countdown.

Best path lengths: **400–1 500 km** · Typical event duration: **15–60 s**

⚙️ HOW IT WORKS

- TX database served from **tx_search.js RAM** (no redundant download) 2.1
- Optional **userlist1.csv** (fmscan.org) enriches TX data with beam directions, PS, PI 2.2
- Fetches live ADS-B positions every **15 seconds**
adsb.one → adsb.lol → adsb.fi (auto-failover)
- Evaluates scatter geometry for every **aircraft × TX pair**
- Applies radio-horizon, ellipse, elevation-angle & **beam direction** gates 2.2
- Ranks results by **composite score 0–100 %**
- Shows top candidates with **ETA countdown** on map
- All external API calls routed via **local Node proxy**

📊 SCATTER SCORE (0–100 %)

Factor	Weight
Sweet Spot (elevation angles) (TX $\sigma=1.2^\circ$, RX $\sigma=3.0^\circ$)	40 pts
Horizon margin ($\sigma \approx 4\,000$ m above envelope)	20 pts
Cross-track distance ($\sigma = 25$ km)	20 pts
Reflection angle (optimum 40° , $\sigma = 25^\circ$)	10 pts
Fuselage alignment ($\perp$ bisector)	5 pts
TX ERP (logarithmic)	3 pts
FM frequency (87.5 vs 108 MHz)	2 pts

× Size multiplier: A5 heavy × **1.30** · A4 large × 1.15 · A3 × 1.00 · A2 × 0.85 · A1 × 0.70

× **Beam multiplier** 2.2 : score reduced to 0 if TX antenna beam does not cover the aircraft position or the RX direction (from userlist1.csv beam data)

Red ≥ 80 % Excellent **Orange ≥ 60 %** High
Yellow ≥ 40 % Medium **Green < 40 %** Low

🕒 ETA COUNTDOWN

Candidates appear **3 min before** and remain **1 min after** the crossing point.

–2:45 Approaching (blue)

NOW ✓ At crossing ± 5 s (green)

+0:38 Past crossing (red)

Dead reckoning fills the gap between 15-second ADS-B updates — icon & ETA refresh *every second* smoothly.

Score forecast evaluates **5 time steps** (0, 60, 120, 180, 300 s ahead); best score is used. Early-exit at score ≥ 95 .

📡 DATA SOURCES

- **TX DB:** tx_search.js RAM
freq, ERP, city, country

📏 GEOMETRY GATES

- TX–RX ≥ 400 km
- Elevation angle to AC $< 9^\circ$ (TX & RX)

🗺️ MAP PANEL

- ● Receiver QTH dot
- ● ● ● TX dots (sized by ERP)

- **userlist1.csv** 2.2
beam, PS, PI, pol — from
fmscan.org (login required)
- **ADS-B:** adsb.one / .lol / .fi
- **Elevation:** opentopodata.org
SRTM 90 m, batch, cached
- **Audio streams:** api.fmlist.org
- **Flags:** flagcdn.com (ITU→ISO)
- **Rotator/GPS:** FM-DX-
Webserver WebSocket
- **Proxy:** local
`/api/airplanesscatter/proxy`

- Both ends inside aircraft's
radio horizon
- $d_{tx} + d_{rx} \leq 1.02-1.08 \times d_{direct}$
- Along-track $\leq d_{txrx} \times 1.1$
- Aircraft > 50 kts & > 1 000 ft
- **TX beam covers AC position** 2.2
- **TX beam covers RX direction** 2.2

- ➔ Aircraft icon (rotated, colour = score)
- - - - Scatter path TX ↔ AC ↔ RX
- Left: candidate list sorted by [ETA]
- TX detail: **PS, PI code, polarisation** from
userlist1.csv 2.2
- Bottom: terrain profile
(interactive, Sweet Spot band)
- ► Audio stream Play/Stop per freq
- Status bar: AC count · candidates · DB size

NEW & CHANGED IN V2.2

- **userlist1.csv integration** 2.2 — optional supplementary database downloaded from fmscan.org (login required). Place `userlist1.csv` in the plugin folder. Provides:
 - Transmitter **beam / radiation direction** (single value or range, e.g. `270-40`)
 - **RDS PS** station name and **RDS PI code**
 - Polarisation, programme name, language
- **Beam-aware scoring & hard gate** 2.2 — if beam data is present in userlist1.csv, the TX antenna beam is checked against both the aircraft position (scatter point) and the RX direction. If either is clearly outside the beam arc, the candidate is suppressed entirely (score forced to 0). Soft rolloff within $\pm 3^\circ$ (range beams) or $\pm 10^\circ$ (fixed beams) ensures smooth transitions at beam edges.
- **Additional TX metadata display** 2.2 — the TX detail panel and map tooltip now show PI code, PS name, polarisation, and beam direction when available from userlist1.csv.
- **Compass filter bug fixed (E / NE)** **FIX** — the E and NE directional sectors were previously using overlapping or incorrect bearing ranges, causing some transmitters to be shown in the wrong compass sector or incorrectly suppressed. Now correctly mapped to $22.5^\circ-67.5^\circ$ (NE) and $67.5^\circ-112.5^\circ$ (E) with proper wrap-around handling for N ($337.5^\circ-360^\circ / 0^\circ-22.5^\circ$).
- **WebSocket connection count fixed** **FIX** — the plugin's own WebSocket connections were previously being counted towards the active client total displayed in the FM-DX-Webserver status, inflating the displayed number. The connections are now correctly excluded from the client count.

TIPS FOR BEST RESULTS

- **Install userlist1.csv** for beam-aware scoring — download from fmscan.org (login required), select *FM+ (Tropo)* → *CSV format* → semicolon separator, download as `userlist1.csv` and place it in `plugins/AirplaneScatter/`. The plugin loads it automatically on startup.
- **Beam data explains missing candidates** — if a high-power TX disappears from the list after installing userlist1.csv, the TX beam is pointing away from you or the aircraft. This is physically correct — scatter cannot occur if the TX antenna doesn't illuminate the reflector.
- **Sweet Spot band** — if the aircraft dot is not inside the green band in the elevation profile, scatter geometry is suboptimal even with a high score.
- **Hover tooltip** — move the mouse over the elevation profile canvas to read terrain height and elevation angles at any path point.
- **Audio Streaming** — start the web radio ~30 s before crossing; streams are typically 15–45 s delayed. Compare audio burst with the stream to positively ID the station.
- Use **AI Denoiser** for weak scatter audio
github.com/Highpoint2000/AI-Denoise
- Use **RDS AI Decoder** for fragmented RDS bursts
github.com/Highpoint2000/RDS-AI-Decoder
- Heavy jets (A380, B747) = **+30 % score bonus** — prioritise!
- Low-power hardware: reduce Fetch Radius to 500 km, raise Min ERP to 100 kW
- Click **↻ Reload** to force a fresh TX database download and clear caches

v2.1a / v2.1 features below ↓

- **Sweet Spot corridor** — green altitude band in elevation profile (TX ≤ 1°, RX ≤ 5°)
- **Profile hover tooltip** — terrain height + elevation angles at cursor
- **TX Envelope Cache** — Float64Array horizon envelopes computed once per RX; ~60–80 % CPU reduction
- **Cheap pre-filter** — Haversine-only gate eliminates ~60 % of pairs before full scoring
- **TX DB from RAM** — shared with tx_search.js, zero download after first restart
- **Audio Streaming** — Play/Stop live web radio per frequency
- **Aircraft Category Filter** — hide light planes (A3+ Large & above)

HOW TO OBTAIN AND INSTALL USERLIST1.CSV (FMSCAN.ORG)

1. Register and log in at fmlist.org (free account required).
2. Navigate to **FMSCAN** → **Userlist** or go directly to fmscan.org/perseus.php?userlistmode=fm.
3. Select: *FM+* (*Tropo*) · CSV format · **Semicolon** separator · Click **DOWNLOAD userlist1.csv** (left-click, wait ~1 minute).
4. Save the downloaded file as `userlist1.csv` and copy it into the `plugins/AirplaneScatter/` folder on your FM-DX-Webserver.
5. Restart the webserver and reload the plugin (↻ button). The plugin will load the file automatically and log the number of entries found.

CSV columns: khz / country / language / program / modulation / location / latitude / longitude / power towards receiver / maximum ERP / main transmitter beam / distance / azimuth / dB / RDS PS / RDS PI / polarisation / ID

⚙ KEY USER SETTINGS (VIA ⚙ BUTTON IN PANEL HEADER)

Setting	Default	What it does
Min TX–RX Distance	400 km	Excludes nearby TX. Scatter needs both ends below radio horizon.
Min TX ERP	100 kW	Filters weak stations. Lower = more candidates, more CPU.
Lead / Trail Time	3:00 / 1:00	Window around crossing during which candidate is displayed.
Fetch Radius	750 km	Radius for TX database & ADS-B aircraft request.
Min Scatter Score	70 %	Minimum peak score required to create a candidate entry.
Aircraft Type Filter	All aircraft	Filter out small aircraft / rotorcraft by ADS-B category (e.g. A3+ Large & above).
Frequency Filter List	None	Load a <code>blacklist.txt</code> or <code>whitelist.txt</code> from the plugin folder to suppress or exclusively show specific frequencies.
RX Antenna Height (AGL)	10 m	Your antenna height above ground — affects horizon & profile.
Unit System	Metric	Switch between km/m and miles/ft/knots.

🔧 RECOMMENDED COMPANION PLUGINS

GPS Plugin — provides live moving QTH coordinates via `/data_plugins` WebSocket. Essential for mobile DXing.
github.com/Highpoint2000/GPS

PST Rotator Plugin — enables Rotor Sync Lock & click-to-turn azimuth. Antenna follows scatter candidates automatically.
github.com/Highpoint2000/PSTRotator

AI Denoiser — neural-network (DTLN/RNNoise) real-time audio cleanup for buried scatter signals.
github.com/Highpoint2000/AI-Denoise

RDS AI Decoder — reconstructs PS station name from just a few corrupted RDS bursts using statistical voting & fmdx.org DB.
github.com/Highpoint2000/RDS-AI-Decoder